



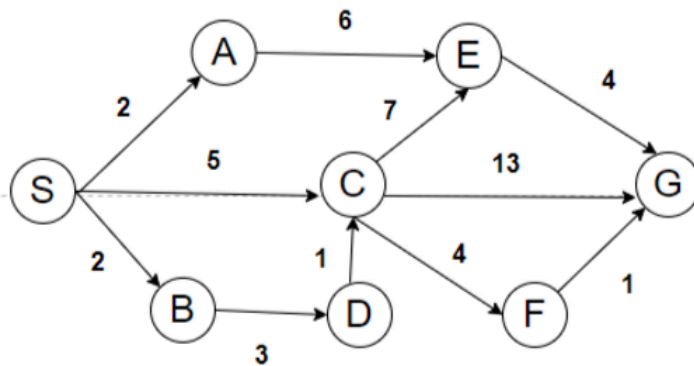
Name:

ID:

Section:

- 1) a. Comment on the admissibility of heuristic and fix any values on the table if needed. (4)
b. Now perform A* search using the edited/updated heuristic values if any. (6)
c. Now comment on the heuristic consistency of the nodes S and F. Must show calculation to justify your answer. (4)
d. Using the updated heuristic table, can we perform a A* Search to find optimal path from state S to node E? Please provide a brief explanation for your answer (2)

states→n	h(n)
S	10
A	8
B	7
C	6
D	7
E	3
F	1
G	0



2) Suppose you have three heuristic functions h_1 , h_2 and h_3 . Among these h_1 and h_2 are admissible but h_3 is inadmissible. You have decided to create several new heuristic functions defined as follows:

- $h_4(n) = 0$
- $h_5(n) = 2 \times h_2(n)$
- $h_6(n) = (h_1(n) + h_2(n)) / 2$
- $h_7(n) = \max(h_1(n), h_2(n))$
- $h_8(n) = \min(h_1(n), h_3(n))$
- $h_9(n) = \max(h_2(n), h_3(n))$

Now answer the following questions and justify. [4]

- Which two heuristics are possibly inadmissible?
- Among h_6 and h_7 which one is dominant?
- In your opinion which heuristic is the best?

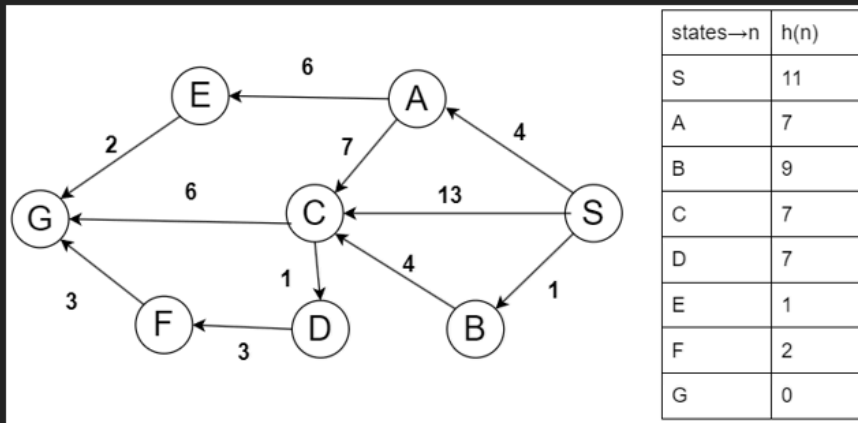


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- b. Now perform A* search on the edited heuristic values if any. (6)
- c. Now comment on the heuristic consistency of the nodes S and F. Must show calculation to justify your answer. (4)
- d. Using the updated heuristic table, can we perform a A* Search to find optimal path from state S to node F? Please provide a brief explanation for your answer (4)



2) Suppose you have three heuristic functions h_1 , h_2 and h_3 . Among these h_1 and h_2 are admissible but h_3 is inadmissible. You have decided to create several new heuristic functions defined as follows:

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- Which two heuristics are possibly inadmissible?
- Among h_6 and h_7 which one is dominant?
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Question-01

You are tasked with designing a seating arrangement for a wedding reception with 8 guests. The guests have specific preferences about whom they would like to sit next to and whom they want to avoid. You need to create a seating arrangement for these guests around a circular table. Depending on the guests' preferred seating arrangements, the aim is to optimize their level of satisfaction. Consider the following constraints:

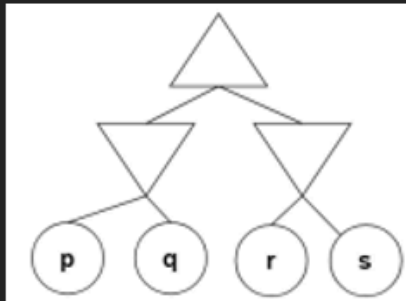
- i. Guest 1 (G1) prefers to sit next to Guest 2 and Guest 3 but wants to avoid Guest 4.
- ii. Guest 2 (G2) prefers to sit next to Guest 1 and Guest 4 but wants to avoid Guest 5.
- iii. Guest 3 (G3) prefers to sit next to Guest 1 and Guest 6 but wants to avoid Guest 7.
- iv. Guest 4 (G4) prefers to sit next to Guest 2 and Guest 5 but wants to avoid Guest 8.
- v. Guest 5 (G5) prefers to sit next to Guest 4 and Guest 6 but wants to avoid Guest 1.
- vi. Guest 6 (G6) prefers to sit next to Guest 3 and Guest 5 but wants to avoid Guest 2.
- vii. Guest 7 (G7) prefers to sit next to Guest 8 but wants to avoid Guest 3.
- viii. Guest 8 (G8) prefers to sit next to Guest 7 but wants to avoid Guest 4.

Answer the following questions based on the scenario above.

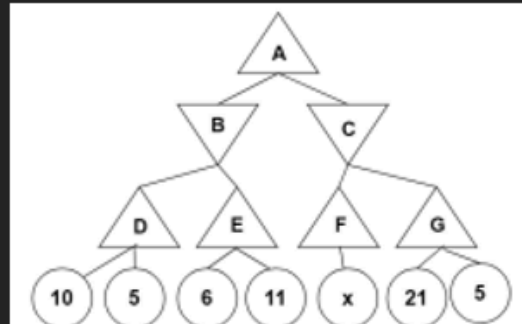
- a. Encode the problem and create **four** parent chromosomes. (3 marks)
- b. Suggest a possible fitness function and choose the two fittest chromosomes. *You may provide answers either in a mathematical form or a descriptive manner.* (4 marks)
- c. Do you think the regular crossover (shown in the n-queen problem) will work in this scenario? If not, explain why (2 marks)
- d. Suggest a possible case of mutation in this context. (1 mark)

Question-02

- 1) Let's say an Agent and Human are playing a game of chess and after 10 turns the Agent won the match. Determine how many times the agent ran Alpha-Beta pruning during the match. Explain in few sentences (2 marks)
- 2) Does ordering of the leaf nodes of a state space tree change the outcome of the MiniMax Algorithm?
Assume p, q, r, s to be some natural numbers which represent the fitness value of the leaf nodes, then show if ordering does affect the outcome of MiniMax algorithm or not. (5 marks)



- 3) For what value of 'x' would the branch from node C to node G gets pruned?





Name :	ID:	Section:
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Question-01

You are tasked with arranging 7 participants for a conference. The conference tables available there are all circular. The participants needed to be seated at the table with maximum satisfaction since some participants prefer to sit next to specific individuals, while others prefer not to be seated next to certain people. Consider the following constraints:

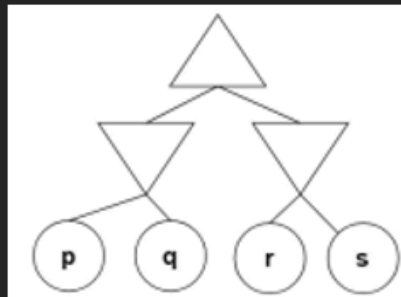
- I. Participant 1 (P1) prefers to sit next to Participant 2 and Participant 3 but dislikes sitting next to Participant 4.
- II. Participant 2 (P2) prefers to sit next to Participant 1 and Participant 4 but dislikes sitting next to Participant 5.
- III. Participant 3 (P3) prefers to sit next to Participant 1 and Participant 5 but dislikes sitting next to Participant 6.
- IV. Participant 4 (P4) prefers to sit next to Participant 2 and Participant 6 but dislikes sitting next to Participant 3.
- V. Participant 5 (P5) prefers to sit next to Participant 3 and Participant 7 but dislikes sitting next to Participant 2.
- VI. Participant 6 (P6) prefers to sit next to Participant 4 and Participant 7 but dislikes sitting next to Participant 1.
- VII. Participant 7 (P7) prefers to sit next to Participant 5 and Participant 6 but dislikes sitting next to Participant 2.

Answer the following questions based on the scenario above. *You may provide answers either in a mathematical form or a descriptive manner.*

- e. Encode the problem and create **four** parent chromosomes. (3 marks)
- f. Suggest a possible fitness function and choose the two fittest chromosomes. *You may provide answers either in a mathematical form or a descriptive manner.* (4 marks)
- g. Do you think the regular crossover (shown in the n-queen problem) will work in this scenario? If not, explain why (2 marks)
- h. Suggest a possible case of mutation in this context. (1 mark)

Question-02

1. Let's say an Agent and Human is playing a game of chess and after 10 turns the Agent lost the match. Determine how many times the agent ran MiniMax Algorithm pruning during the match. Explain in few sentences(2 marks)
2. Does ordering of the leaf nodes of a state space tree change the outcome of Alpha Beta pruning Algorithm?
Assume p, q, r, s to be some natural numbers which represent the fitness value of the leaf nodes, then show if ordering does affect the outcome of Alpha Beta algorithm or not. (5 marks)



3. For what value of 'x' would the branch from node C to node G get pruned?

